

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

Chorok Lee | KAIST

A model can be confidently wrong after the world changes. Conformal prediction adds a set of plausible answers, but even that set can stop containing the truth. This paper asks whether the model's feature reliance offers an early warning before future data arrive.

The question

Coverage is the fraction of examples whose true label appears in a prediction set. Standard conformal guarantees require exchangeability between calibration and future examples. The study examines why classifiers facing a shared COVID-era time split can have very different coverage losses. Its focus is gradient-boosted multiclass classifiers with Adaptive Prediction Sets (APS), targeting 90% coverage.

The approach

The proposed diagnostic is top-1 SHAP concentration: the largest mean absolute feature attribution divided by the sum across features. It uses validation data only. If one feature dominates the model's measured reliance, a damaging change affecting that feature may make the predictor fragile. An idealized probability-mixture theorem formalizes a mechanism under explicit assumptions; it is not a guarantee for measured TreeSHAP concentration.

What the evidence shows

Eight SALT supply-chain tasks exhibit coverage losses from 0.1 to 77.1 percentage points. The main experiments use 50 independent training seeds per task, not a 50-model prediction ensemble. Across a pooled set of 16 multiclass tasks in 9 domains, concentration and coverage loss have Spearman correlation 0.853, with reported $p < 0.001$ and bootstrap 95% interval [0.50, 0.96]. The pool comprises the original eight SALT tasks and eight external tasks. Four external tasks have documented shifts and four are null-shift controls. Within SALT, standard shift indicators are weakly associated with the severity of coverage loss.

What the result does not promise

The exploratory 40% cutoff has exceptions. Sales-office exceeds it but barely loses coverage; KDDCup99 falls below it but has a seed-sensitive mean loss above the study's at-risk cutoff. A protective-secondary-feature rule is based on one case. The diagnostic is weaker for random forests and MLPs; low concentration does not rule out neural-network failures. The pooled correlation is not wholly independent validation of the threshold, and an association does not show that reducing concentration will necessarily improve coverage.

Why it matters

The work adds a model-specific question to reliability assessment: does a plausible shift threaten the features on which this predictor relies? Concentration can guide inspection and monitoring before deployment. A prospective evaluation should freeze the diagnostic rule and test both coverage and prediction-set usefulness. Adaptive calibration may recover coverage by returning larger sets, so apparent statistical recovery can still leave a system less useful.

KEY TAKEAWAY

Treat concentrated reliance as a warning to investigate, especially for boosted multiclass conformal models. It is not a certificate of failure or safety.